

NFL ELO

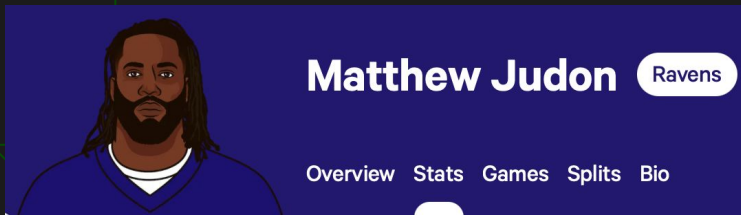
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Outline



- Current player evaluation background
- Intro to ELO Modeling
- Our model + results
- Conclusion

Problem – Who is a better player?



Matthew Judon Ravens

Overview Stats Games Splits Bio

Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	NE	17	38	22	60	14	12.5



Aaron Donald Rams

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Current evaluation methods

1. Pass block win rate
Proportion of attempts in which pass rusher beats their blocker within 2.5 seconds
2. Aggregated statistics
Simple counts of sacks, hits, or hurries or QB statistics
3. Film evaluation
Viewing offensive line or rusher play through playback



Issues With Current Evaluation Methods

No Adjustment for Match Up Difficulty

Opposition player quality or double teams not taken into account

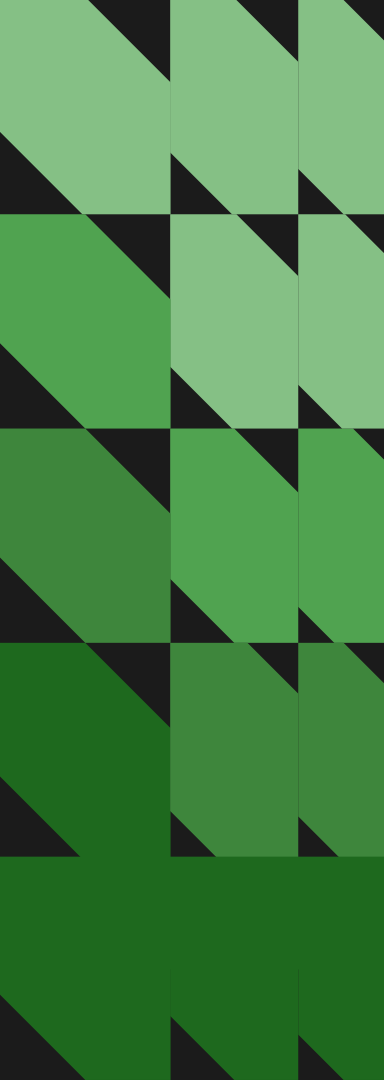
No Differentiation for Impact of Event

Hits, hurries, and sacks each have differing impacts on the game

Measures are not Dynamic

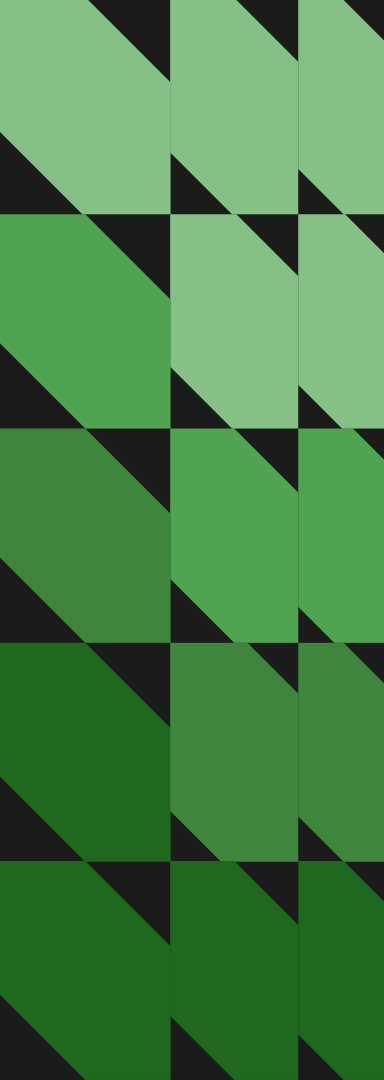
Aggregated statistics are insensitive to within season change and are not sequential





Data - Hudl 2021 Season

- Records game freeze frames with a granularity of 0.1 seconds, time between events labeled pathing with empty/NA rows
- Records locations, timings and players involved in events
- Labeled defensive/offensive line matchups



Encoding Results of Plays

- **Double team**
 - overlap between blocker engagements on the same rusher
- **Beat**
 - Did rusher get closer to the QB than the blocker before the play ends or a pass event occurs
- **Hit**
 - Tackle on a quarterback before the pass and behind line of scrimmage that isn't a sack
- **Sack**
 - Already defined as event in data

The ELO model

Formula for Updating Scores

$$R'_A = R_A + K(S_A - E_A)$$

- R'_A : New Elo score of player A
- R_A : Original Elo score of player A
- K : Learning rate parameter
- S_A : Score of the event for A
- E_A : The expected score for A

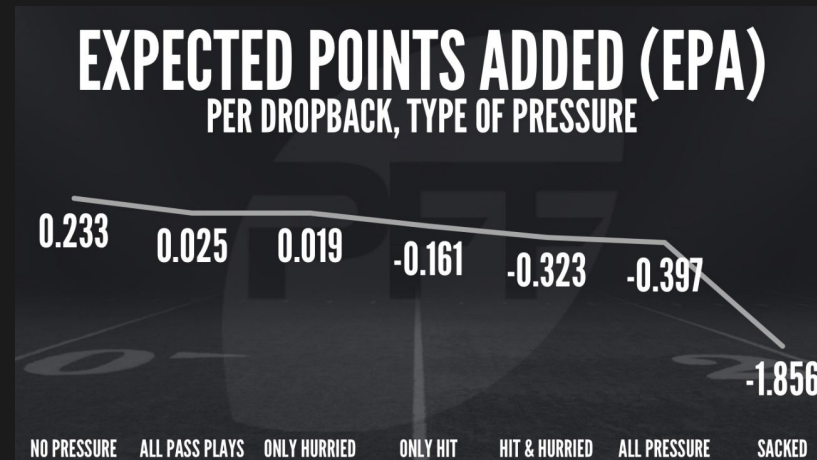
Obtaining Win Probabilities from Elo

$$P(\text{A wins}) = \frac{1}{1 + 10^{(R_B - R_A)/\alpha}}$$

- R_A : rating of player A before matchup
- R_B : rating of player B before matchup
- α : Scaling parameter (conventionally 400 in chess)

Multinomial ELO

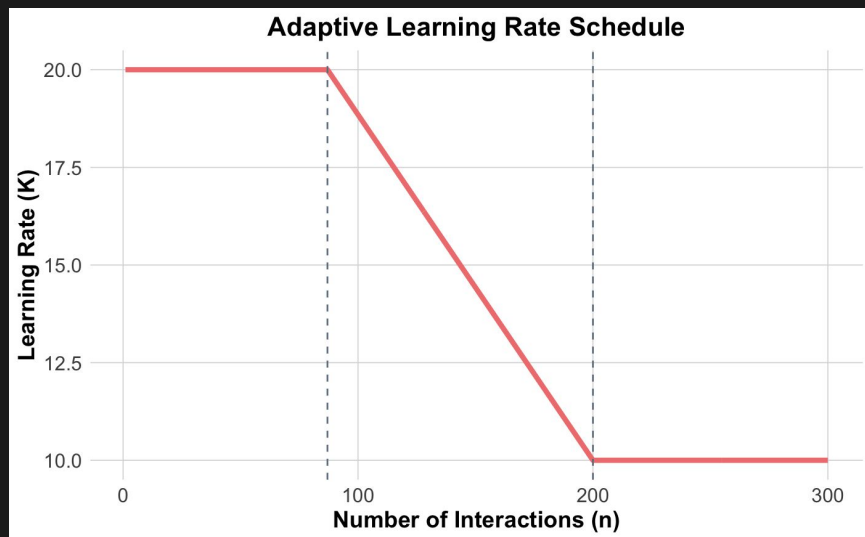
- Lack of pressure, beats, hits and sacks all have different effects on the expected points in a game (analysis by Eric Eager)
- Sacks with outsized impact compared to other outcomes

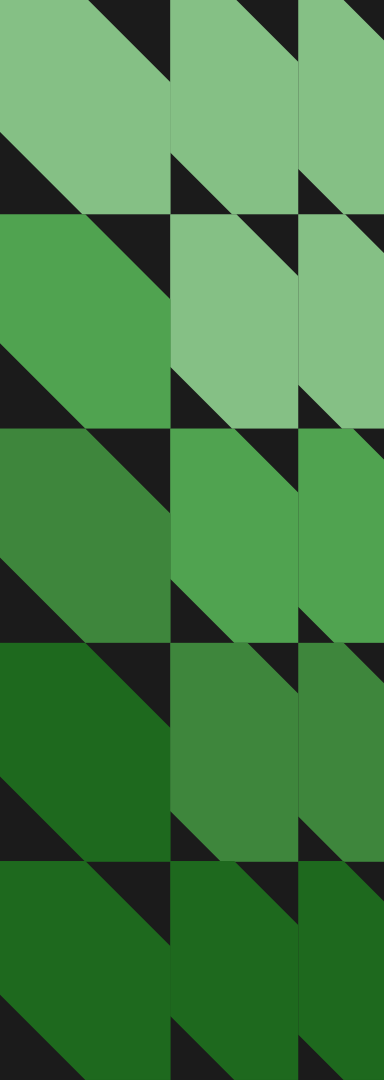


We weigh the result of the pass rush by the respective impact in our model

Adaptive learning rate

$$K(n) = \begin{cases} K_{\text{start}}, & \text{if } n \leq n_{\text{prov}} \\ K_{\text{start}} + (K_{\text{min}} - K_{\text{start}}) \cdot \frac{n - n_{\text{prov}}}{n_{\text{decay}} - n_{\text{prov}}}, & \text{if } n_{\text{prov}} < n \leq n_{\text{decay}} \\ K_{\text{min}}, & \text{if } n > n_{\text{decay}} \end{cases}$$





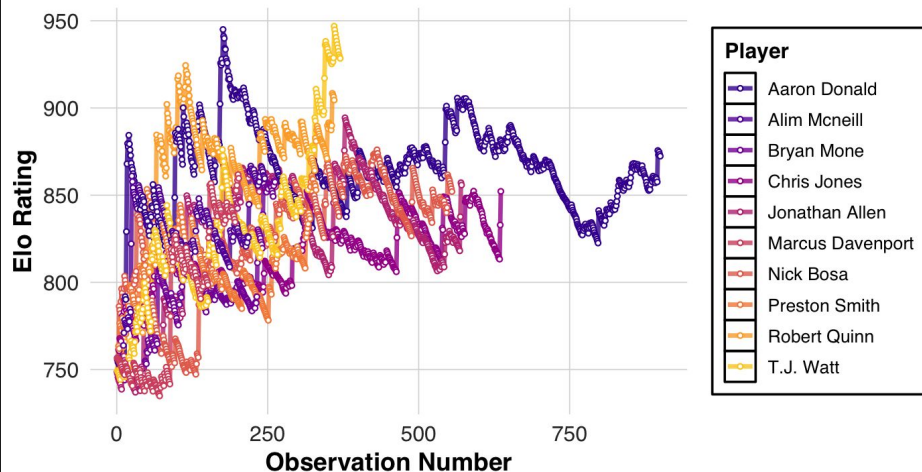
Other Considerations

- Double teams
 - Much harder for the pass rusher to win in these cases
 - Group by pass rusher and identify blocks that overlap as double teams
 - Add a 100 ELO bonus to the blockers in this case -> reflect the halved probability of winning a double team
- Hyperparameters
 - Scale = 319
 - Starting ELOs 1150 and 750 for blockers and rushers, respectively

Results – 2021 Full Season

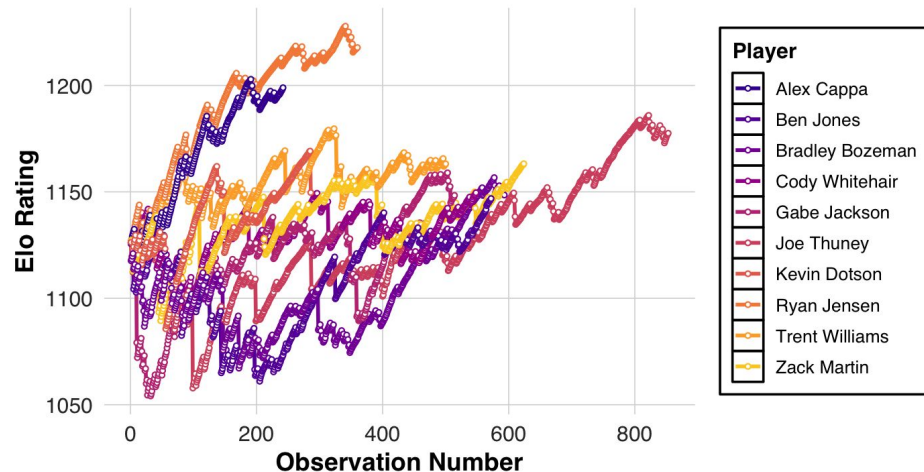
Top 10 Rushers: ELO Trajectories

Evolution of Elo rating over successive plays



Top 10 Blockers: ELO Trajectories

Evolution of Elo rating over successive plays



- Filtered for >200 interactions
 - One double team = 2 interactions for the rusher
- We see that the best players have an increase relative to their starting ELOs

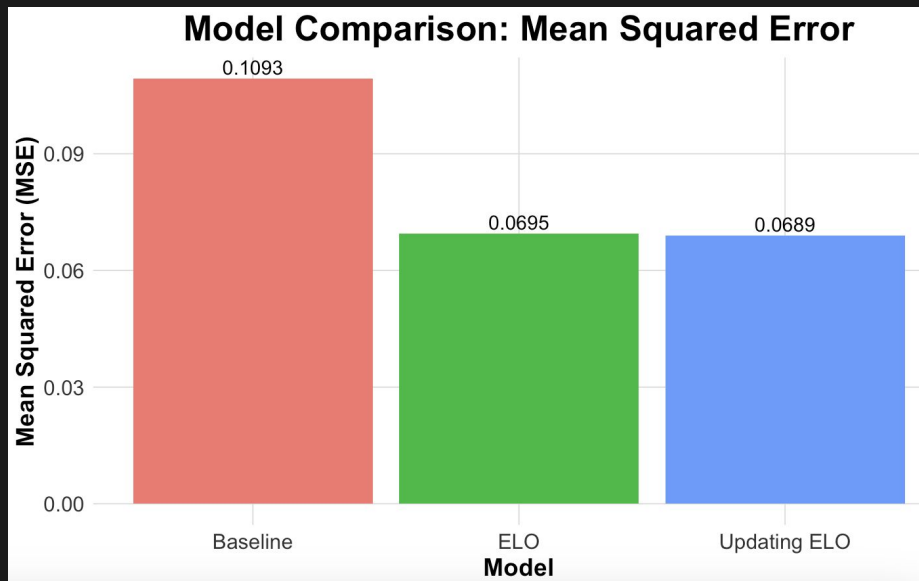
Model Validation

Split the data into train and test (80/20)

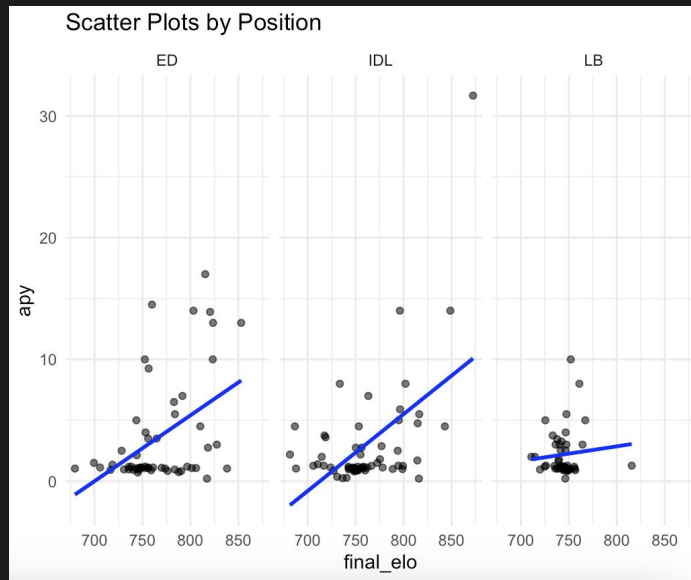
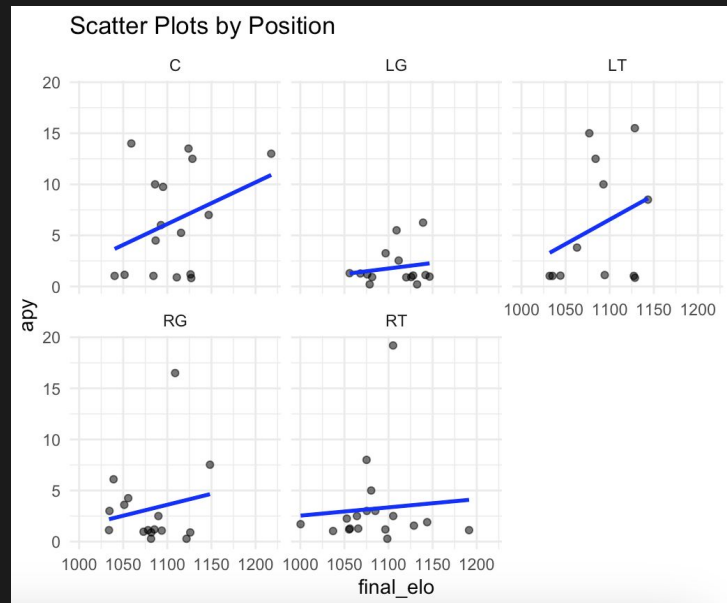
Baseline metric: average win rate observed in train

Elo: take final elos of players in train, use those for prediction

Updating Elo: using previous elo score for prediction



Plotting Elos against Player Pay

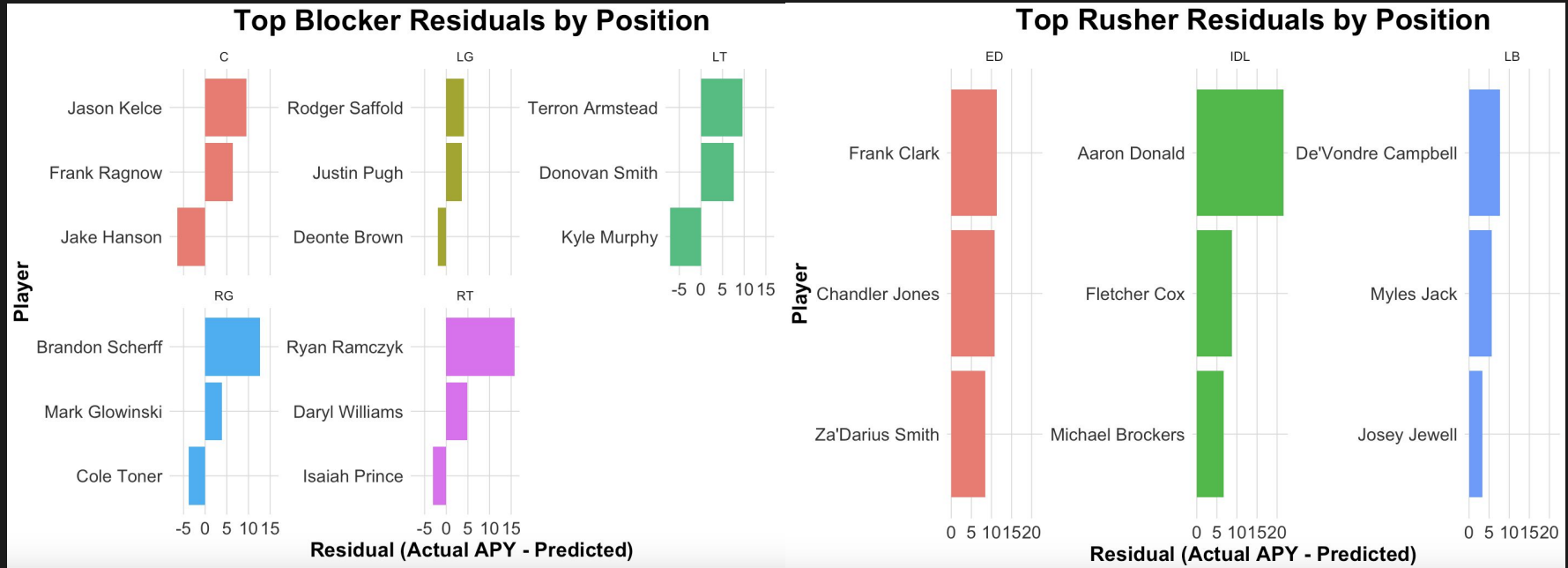


Annual pay of contracts signed in 2021 and 2022 (Millions) against final 2021 ELO (rookie contracts excluded)

What would we want to see?

Positive slope since a higher ELO should be a better player who should be getting paid more (on average)

Player salary outliers



- We see that the largest rusher residuals are purely positive
 - Probably indicative of the nonlinear relationship between performance and salary for top rushers

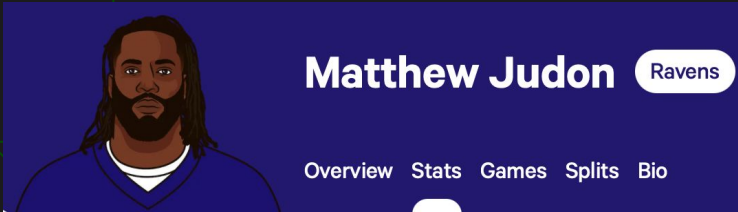
Challenges + Next Steps

- Could not account for penalties
 - Only NA present as penalty type
- Possibly center ELO at 0 for greater interpretability
- Also look at other methods of visualization
 - Week by week trajectories
- Run model on multiple seasons
- Is there a way to calculate ELO for run plays?



Who is a better player?

Elo = 793



Matthew Judon Ravens

Overview Stats Games Splits Bio

Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	NE	17	38	22	60	14	12.5

Prob of beating average blocker: 10.3%

Elo = 872



Aaron Donald Rams

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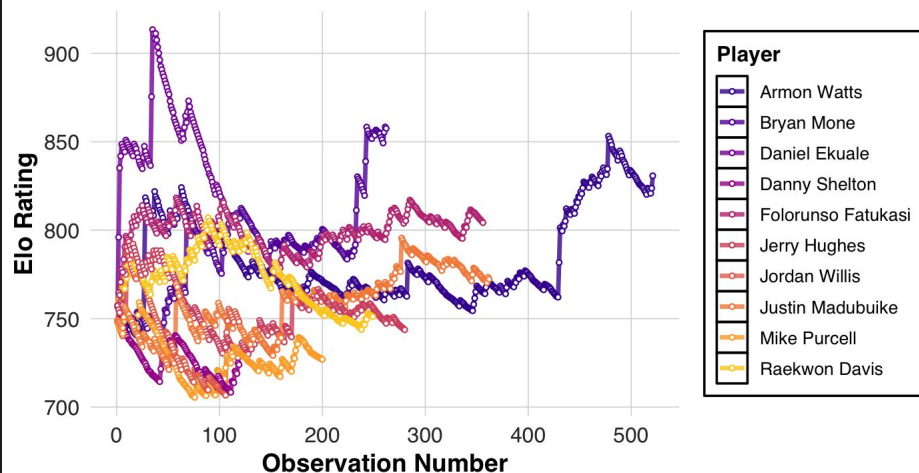
Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	LAR	17	38	46	84	19	12.5

Prob of beating average blocker: 16.9%

Random sample of 20 players

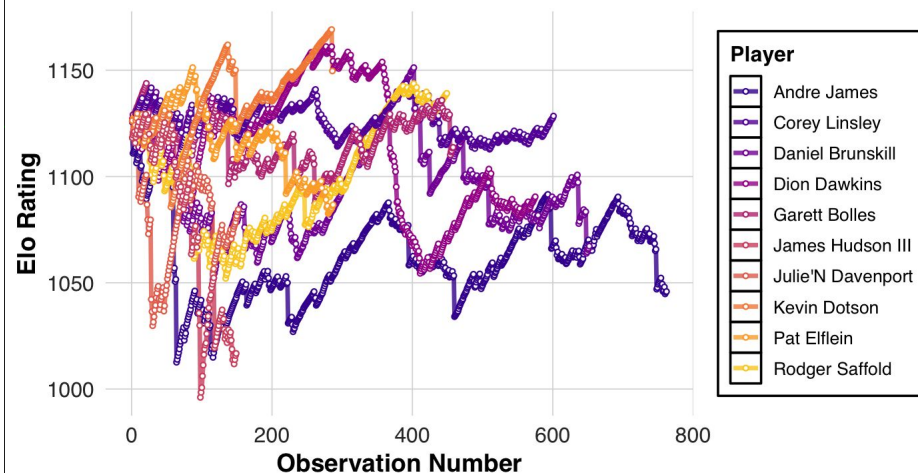
Random 10 Rushers: ELO Trajectories

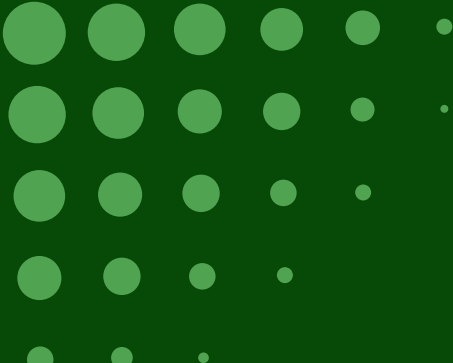
Evolution of Elo rating over successive plays



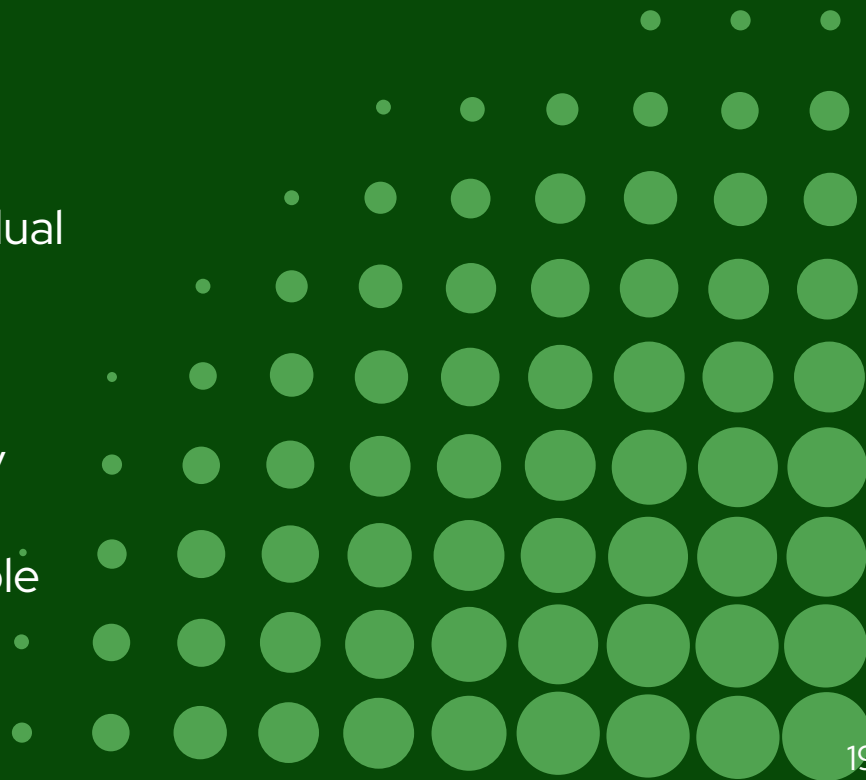
Random 10 Blockers: Elo Trajectories

Evolution of Elo rating over successive plays





Our research questions

1. How can matchup difficulty at the individual level be incorporated into pass rusher evaluation?
 2. How can performance on a play-by-play basis be dynamically assessed to identify temporal shifts in skill?
 3. Which evaluation metrics are most suitable for comparing play-level approaches vs traditional, static models?
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Pass Rusher Sniff Test

Rushers	Elo	Sacks	Tackles	Games Played
B.J. Hill	889	5.5	50	16
Chris Jones	892	9	27	14
Denrico Autry	890	9	31	17
Jacob Martin	898	4	23	17
Joey Bosa	879	8	33	12
Nick Bosa	896	15.5	52	17
Robert Quinn	919	18.5	49	16
T.J. Watt	867	22.5	64	17
Trey Hendrickson	874	14	34	16
Yannick Ngakoue	894	10	28	17